Letters to the Editor

Citation searching power

Dear Sir,

The article on chemical toxicology searching by D. Bawden and A.M. Brock [1] is a valuable study, both for the light it sheds on information retrieval in that area, and for its useful discussion of the methodology used.

One aim of the project was to reach "an objective assessment of the strengths and weaknesses... of searching techniques" through an evaluative procedure which "enabled retrieval failures due to sub-optimal use of a particular source... to be identified." It is therefore surprising that, whilst SCISEARCH was used and in some queries yielded unique references, none of the participants used citation searching: neglecting to use a major search technique in a source acknowledged to contain information not retrieved elsewhere is surely sub-optimal (particularly when one finding of the study is that a higher rate of retrieval can result from the provision of additional access points).

This is clearly not the place for a full discussion of citation indexing. However, whilst one answer to the difficulties reported – on the one hand with highly specific indexing, and on the other with lack of indexing of substance or toxicity – would be a specialised data bank with a formalised vocabulary for encoding toxic effects; some of the indexing problems encountered could almost certainly have been overcome by the use of citation searching, with its potential for great depth of indexing and its avoidance of many of the semantic problems inherent in traditional indexing systems.

Yours faithfully

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Reference

[1] D. Bawden and A.M. Brock, Chemical toxicology searching: a collaborative evaluation, comparing information resources and searching techniques, *J. Information Sci.* 5(1) (1982) 3–18.

The vicious circle of citation

Dear Sir,

The juxtaposing of the articles by Michel and Lawani (*Journal of Information Science 5* (4) 131–135 and 143–145) subtly highlighted the problems involved in identifying quality in scientific performance.

Lawani, in the methodology section of his paper on the relationship between the quality and quantity of research productivity, states that "... the panel's quality judgments are valid and dependable considering *the known positive correlations between quality and citation*" (italics added). The author supports his statement by references to four articles which have identified correlations between citedness and other recognised quality indicators. He does not, however, refer to the dissenting literature, and the question of construct validity is sidestepped.

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Michel, meanwhile, draws our attention to the vicious circle of citation:

"– The best publications are the most cited articles.

– The most cited publications are articles which are most read.

– English is the most common language in the scientific community. In order to be cited, it is better to publish in English. Therefore, publications in English are probably the best ones and will have to receive top priority in selection."

If citation indexes selectively favour English language publications (and, by extension, those who publish in English), then global quality assessments based on citation counts will reflect and perpetuate this bias.

A recent paper which takes issue with citation numerology, which your readers may wish to refer to, is Matter and Broms' "The myth of Garfield and citation indexing". This was published in *Tidskrist för Dokumentation 39*(1), (1983) 1–8, and is, therefore, unlikely to achieve the readership it warrants. The authors' point is that new schools of thought, particularly in the Social Sciences and Humanities (e.g. phenomenology, structuralism, psychoanalysis) are not properly identifiable via citation indexes. This, the authors maintain, is especially true of much European research. They illustrate their point by looking at, *inter alia*, the case of de Saussure, who, though an internationally recognised thinker, was cited a meagre 25 times in SSCI. Garfieldian indexes, they argue, place the accent on in-vogue research and on American normal science in particular. This, of course, has unfortunate implications for scientists and researchers publishing in languages other than English—the point being made by Michel.

Evaluating the quality of scientific research (whether in the Natural or Social Sciences) is a difficult undertaking. Michel's point seems to be that the (now almost institutionalised) reliance on citation is leading to the creation of "a kind of English speaking community ghetto". Future analyses of research quality should take note of this fact.

Yours sincerely,

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